

Do practical GKP preparation protocols follow the time-optimal path?

Working draft — internal

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Abstract

Preparation of finite-energy GKP states is judged almost entirely by fidelity. Fidelity certifies arrival; it says nothing about the *route*. For this target the time-optimal route is known in closed form: the Fubini–Study geodesic generated by the quantum brachistochrone Hamiltonian H_B , which sweeps vacuum into $|+Z_L\rangle$ along a clean, symmetric path. Practical preparation protocols are built from a different set of concerns — what a device can actually apply — and there is no reason they should resemble that path. We introduce D_{path} , the mean distance from a trajectory to the geodesic curve, distinguish it from the path-length ratio $\mathcal{R}_{\text{length}}$, and calibrate both against a random-state null. Applying these to continuous pump pulses and to hardware-native gate sequences (displacement+SNAP, echoed conditional displacement), we find that gate sequences reach high fidelity while travelling through states nearly orthogonal to the entire geodesic, and that a least-squares analysis pins the cause: H_B lies outside the span of the available control operators, so only 62–79% of the required velocity can ever be produced. Optimizing both families explicitly against the geodesic settles the comparison on equal footing: pulses reach $D_{\text{path}} \approx 0.16$ at $F = 0.99$ where gate sequences plateau near 0.76, a gap *wider* than a fidelity-only comparison suggests. Along the way we retract a claim of our own — an apparent geodesic “floor” for pulses was slack, removable for 0.8% fidelity — and identify a degeneracy that any path-deviation objective must guard against.

1 Motivation

The analytic layer of this project supplies something unusual: an explicit time-evolution Hamiltonian that drives vacuum to a finite-energy GKP logical state along the time-optimal path. Writing $|\psi_i\rangle$ for vacuum and $|\psi_f\rangle = |+Z_L\rangle$, the brachistochrone construction gives a constant H_B acting in $\text{span}\{|\psi_i\rangle, |\psi_f\rangle\}$ whose flow traces the Fubini–Study geodesic and saturates the Mandelstam–Tamm speed limit. It is a reference object, not an implementable one, but it fixes what an *optimal* preparation looks like — including that the intermediate states form a clean, symmetric family interpolating vacuum and the target lattice.

Practical GKP preparation looks nothing like this. The protocols that hardware can run — displacement interleaved with number-selective phase gates, or echoed conditional displacements against a transmon ancilla — are assembled from device-native primitives, and reach the target by routes chosen for implementability rather than optimality. The question this document addresses is how far apart those two things are, and whether the gap is a property of the protocols or merely of how we optimize them.

Our contributions are: (i) a metric, D_{path} , that measures deviation from the time-optimal *path* rather than from the target state, validated against the brachistochrone flow itself; (ii) measurements of that metric across continuous and gate-based protocols; (iii) a least-squares diagnostic that identifies the structural obstruction; and (iv) an explicit account of which of our own claims survived scrutiny and which did not.

2 Setup

2.1 Target and control families

The target is the finite-energy logical state $|+Z_L\rangle$ at envelope width $\Delta = 0.3$, built as a lattice sum of displaced vacua with a Gaussian finite-energy envelope, truncated at $n_{\text{fock}} = 100$. It carries $\langle n \rangle = 5.83$ and has 99.9% of its population below Fock level $N_{\text{phys}} = 42$.

Quantity	Definition	Question it answers
$\mathcal{R}_{\text{length}}$	L/θ	Did the trajectory take a <i>detour</i> ?
D_{path}	$\langle \min_s \arccos \langle \psi(t) \gamma(s) \rangle \rangle_t$	Did it stay on the <i>route</i> ?
$\mathcal{R}_T$	$T/T_B, T_B = \theta/\Delta H_{\text{max}}$	How much slower than the speed limit?

Table 1: The three geometry metrics. $\mathcal{R}_{\text{length}}$ and D_{path} are independent: a trajectory can be short yet bow far off the geodesic, and Section 5.5 exhibits a regime where they move in opposite directions.

Two control families are studied.

Continuous pumps. The even-photon pump hierarchy

$$X_n = (a^\dagger)^n + a^n, \quad Y_n = i(a^n - (a^\dagger)^n), \quad n \in \{2, 4, 6, 8\}, \quad (1)$$

driven with time-dependent real amplitudes $u_{n,x}(t), u_{n,y}(t)$. Parity excludes odd orders: vacuum and $|+Z_L\rangle$ are both even under photon-number parity.

Hardware gates. Displacement $D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a)$, interleaved with the number-selective phase gate $\text{SNAP}(\vec{\theta}) = \sum_n e^{i\theta_n} |n\rangle \langle n|$; and echoed conditional displacement

$$\text{ECD}(\beta) = \exp\left[\frac{1}{2}(\beta a^\dagger - \beta^* a) \otimes \sigma_z\right], \quad (2)$$

interleaved with qubit rotations, acting on the joint cavity–transmon space. ECD is ancilla-mediated: cavity and transmon entangle during the sequence and are disentangled only at the end.

Operator normalization.. Raw pump norms scale as $n_{\text{fock}}^{n/2}$, spanning 1.8×10^2 to 1.0×10^8 at $n_{\text{fock}} = 100$; quasi-Newton optimizers abort on the first line search. Normalizing by the *full* spectral norm fixes that but is dominated by the truncation edge, leaving X_8 roughly $19\times$ more weakly coupled than X_2 where the states actually live. We therefore normalize each operator by its norm restricted to the populated block, $\|O_{[0:N_{\text{phys}}]}\|_2$.

2.2 The geodesic reference

With $c_0 = |\langle \psi_i | \psi_f \rangle|$ and $\theta = \arccos c_0$, the geodesic is the great-circle arc in the two-dimensional subspace spanned by $|\psi_i\rangle$ and the component of $|\psi_f\rangle$ orthogonal to it,

$$|\gamma(s)\rangle = \cos s |\psi_i\rangle + e^{-i\phi} \sin s |\psi_f^\perp\rangle, \quad s \in [0, \theta], \quad \phi = \arg \langle \psi_i | \psi_f \rangle, \quad (3)$$

parameterized by Fubini–Study arc length. For our target $c_0 = 0.5442$ and $\theta = 0.9953$ rad. Equation (3) is exactly the trajectory generated by H_B .

3 Metrics

By the Anandan–Aharonov relation a state evolving under $H(t)$ moves through ray space at speed equal to its energy uncertainty, $ds_{\text{FS}}/dt = \Delta H(t)$ ($\hbar = 1$), so the path length is $L = \int \Delta H dt$. From this we define

D_{path} takes the distance to the *nearest* point of the geodesic, not to the same-time point, so it is invariant to how fast the path is traversed: a protocol that follows the geodesic slowly still scores zero.

Calibration.. A value in radians is uninterpretable without endpoints. The geodesic itself returns $D_{\text{path}} = 9 \times 10^{-9}$ and $\mathcal{R}_{\text{length}} = 1.000000$ (feeding H_B in as a single gate), confirming the implementation. Random states drawn in the populated subspace sit at $D_{\text{path}} = 1.395$; the hard maximum is $\pi/2 = 1.571$, attained only if every trajectory point is orthogonal to the whole geodesic. The random-state null is the meaningful upper reference: it is what a trajectory scores for ignoring the geodesic entirely.

A caveat that invalidates naive comparisons.. At low fidelity D_{path} is small simply because the state never travels far — a trajectory that stops near $\gamma(0)$ cannot deviate. Comparisons must therefore be made at matched fidelity. This artifact invalidated three separate comparisons during this work before it was identified, and it is the reason several tables below carry fidelity alongside geometry.

Non-circularity.. Except where explicitly stated (Section 5.5), the geodesic never enters an optimization objective. Optimizers see fidelity, and geometry is computed afterwards. Without this discipline any statement that a protocol “approaches time-optimality” is circular.

4 How the protocols are optimized

Continuous pulses.. Amplitudes are parameterized by band-limited Fourier coefficients and optimized by L-BFGS-B on analytic gradients obtained by differentiating through the propagator product $\prod_k \exp(-i \Delta t H(u_k))$. This is GRAPE in the usual sense.

Gate sequences — not GRAPE.. For a gate sequence the free parameters are not time-slice amplitudes but the *gate parameters*: the complex displacements α_k and the phase vectors $\vec{\theta}_k$ (or, for ECD, the β_k and qubit rotation angles). The sequence *structure* — how many layers, and the fixed alternation — is held fixed, and only the continuous parameters are fitted, by multi-start L-BFGS-B differentiating through the gate exponentials. So the optimizer family is the same quasi-Newton gradient method; what differs is the parameterization. There is no time discretization, and no notion of a control amplitude per slice.

This split is deliberate: the continuous parameters are exactly what gradients handle well, while the discrete structure — which gate, how many, in what order — is combinatorial and is the part a learned or search-based method would have to supply.

The Gaussian ceiling.. Direct optimization from a small-amplitude random start stalls at $F = 0.573$ for every method we tried, including reinforcement learning. This is not an optimizer artifact but a statement about the accessible states: optimizing over *all* pure Gaussian states reachable from vacuum — displacement and squeezing together, four free real parameters — gives a maximum overlap with the target of 0.5729, matching the stall exactly. Of the eight pump controls only X_2 and Y_2 are quadratic in a and hence Gaussian; X_4 through Y_8 are the non-Gaussian resource. Getting past $F = 0.573$ therefore *is* the act of engaging non-Gaussianity, and since GKP states are Wigner-negative — a property no Gaussian operation can create — no protocol can succeed without it.

We verified the stall is a genuine local optimum rather than a saddle or a tolerance artifact: at the stalled point the gradient norm is 5×10^{-10} , the gains are $(0, 43.2, 0, \dots, 0)$ (pure squeezing), and 200 random directions at each of six step sizes spanning 10^{-3} to 10^2 are all downhill. The basin is wide — perturb-and-reoptimize escapes only for perturbations exceeding roughly twice the solution’s own norm — which is why small-amplitude multistarts land in it reliably.

Escaping it.. We use homotopy continuation in the envelope width, annealing Δ from 1.0 to 0.3 and warm-starting each step: at large Δ the target is nearly Gaussian and squeezing almost suffices, and each small decrease keeps the previous solution inside the new basin. This is continuation in *degree of non-Gaussianity*. The resulting constant-Hamiltonian solution then warm-starts the time-dependent optimization.

It is not the only route out. Large-amplitude random restarts also escape, and on raw fidelity did better ($F = 0.877$ against the curriculum’s 0.776 for the same eight-pump constant-Hamiltonian problem). What continuation appears to buy is not fidelity but *geometry*: its solution has $\mathcal{R}_{\text{length}} = 56$, where a large-random-init solution at equal final fidelity reaches $\mathcal{R}_{\text{length}} = 17079$. Escaping the basin and escaping it well are different things — a distinction that recurs in Section 5.5.

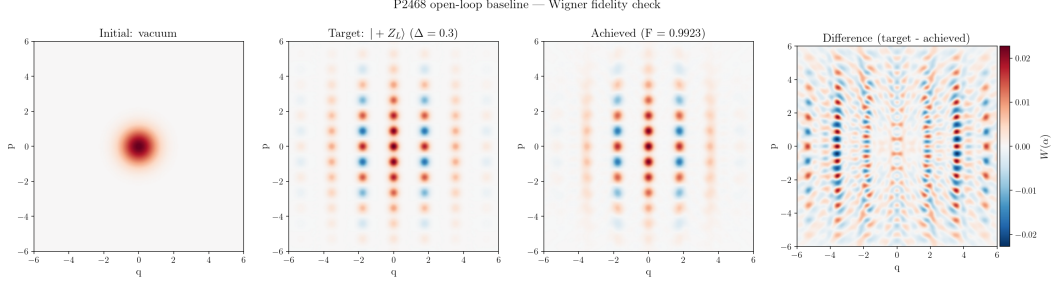


Figure 1: Wigner functions for the continuous-pulse baseline. The achieved state is visually indistinguishable from the target; the difference panel peaks at ± 0.02 against grid features of ± 0.3 , with the residual confined to the outer, high-photon-number peaks.

Variation	Range covered	Effect on D_{path}
SNAP width (gate-set richness)	$2 \rightarrow 42$ phases ($21\times$)	$r = -0.09$ (none)
Gate algebra	displacement+SNAP vs. ECD	Welch $p = 0.71$ (none)
Ancilla use	none vs. maximal entanglement ($\ln 2$)	$r = -0.12$ (none)
Circuit depth	$3 \rightarrow 16$ layers	weak
Fidelity	$0.56 \rightarrow 0.999$	weak

Table 2: Invariance of D_{path} across 26 gate configurations (1.031 ± 0.089). We predicted ECD would sit *further* out, since ancilla-mediated control routes through entangled states nearly orthogonal to the product-state geodesic. ECD does reach maximal cavity entropy, and D_{path} does not move: entanglement is not the mechanism.

5 Results

5.1 Continuous pulses reach the target; the path is another matter

The eight-pump family alone reaches $F = 0.992$ at $T = 2 \mu\text{s}$, matching the existing static baseline of 0.984; adding detuning and Kerr as controls changes nothing once the schedule is time-dependent. Figure 1 confirms the fidelity is real rather than an artifact of a poorly-conditioned overlap.

5.2 Minimum time shortens the path without straightening it

Sweeping the horizon at fixed target (Figure 2), $\mathcal{R}_{\text{length}}$ changes by $25\times$ across the sweep while D_{path} changes by only $1.35\times$. Time pressure compresses the route; it does not move it onto the geodesic.

5.3 Gate sequences are twice as far out, and nothing we varied moved them

Table 2 summarizes 26 gate configurations. D_{path} is 1.031 ± 0.089 throughout — against 0.46 for pulses and 1.395 for random states. On the scale from geodesic to random, gate trajectories are 74% of the way to random; pulses are 32%.

5.4 The structural obstruction, measured directly

Staying on the geodesic requires matching its velocity. Projecting out the unphysical component along $|\psi\rangle$, this is a linear least-squares problem in the real amplitudes at every timestep,

$$\min_u \left\| P \left(H_{\text{drift}} + \sum_k u_k O_k \right) |\psi\rangle - P H_B |\psi\rangle \right\|, \quad P = \mathbb{I} - |\psi\rangle \langle \psi|, \quad (4)$$

solvable analytically — no optimizer, no seeds, no local minima. The pump family realizes only 62–79% of the required velocity direction: H_B is genuinely not in the span of the control operators. The resulting tracker *cannot prepare the state*: fidelity peaks at 0.724 and then degrades as it drifts

Minimum-time GKP preparation: a shorter path is not a straighter one

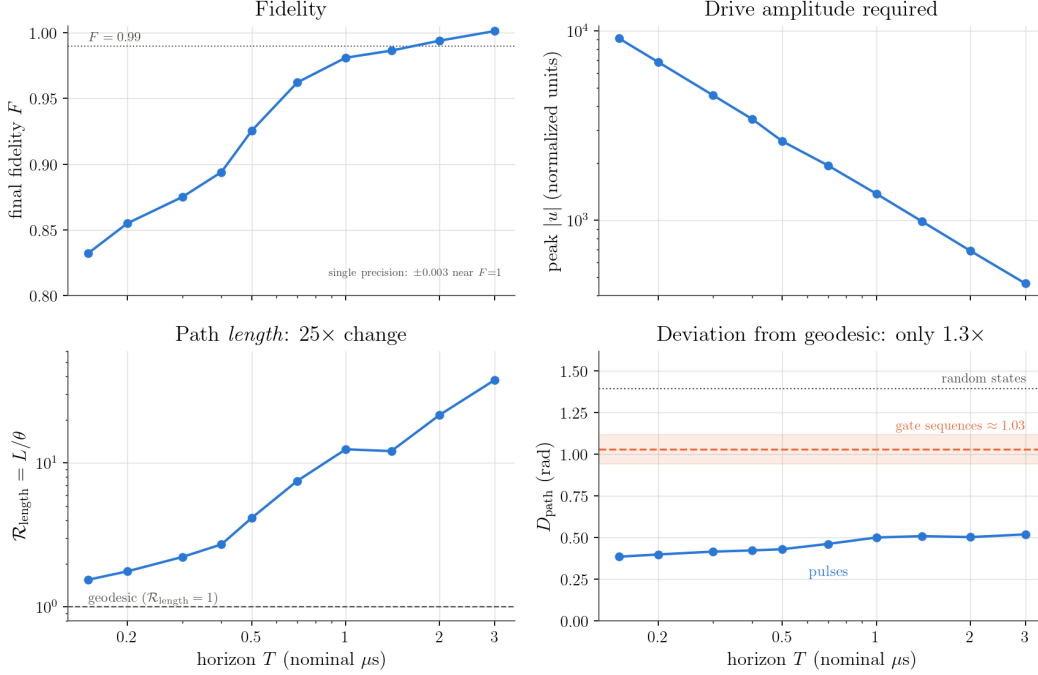


Figure 2: Horizon sweep for the pump family. Path *length* collapses by $25\times$ as T shrinks; deviation from the geodesic moves by $1.35\times$. Efficiency is bought with amplitude.

and overshoots. This is the most robust statement in this work, because it concerns operator spans rather than what an optimizer happened to find.

5.5 A claim of ours that did not survive

Every D_{path} above was measured under a fidelity-only objective. We had concluded that $D_{\text{path}} \approx 0.45$ was the pump family’s intrinsic distance from time-optimality. Putting the geodesic *into* the objective and sweeping the weight,

$$\mathcal{L} = -F + \lambda D_{\text{path}}, \quad (5)$$

refutes that (Table 3, Figure 3). The family reaches $D_{\text{path}} = 0.155$ at $F = 0.992$: three times closer for eight tenths of a percent of fidelity. The 0.459 was slack, not structure. Note also that $\mathcal{R}_{\text{length}}$ *increases* as D_{path} falls — hugging the curve costs a longer path — which is direct evidence the two metrics are not proxies for one another.

This experiment places the geodesic in the objective by construction. It therefore measures a *reachability limit*, and must not be conflated with the minimum-time results, which keep the geodesic in evaluation and support the separate claim about what time pressure alone produces.

5.6 D_{path} has a trivial minimizer

Any objective containing D_{path} must contend with a degenerate solution: vacuum *is* $|\gamma(0)\rangle$, so a trajectory that does nothing at all scores $D_{\text{path}} = 0$ exactly, at $F = c_0^2 = 0.2962$. For $\lambda \gtrsim 0.5$ this trivial point beats any honest attempt on the raw objective, and a multi-start optimizer falls straight into it — as ours did, returning $F = 0.2962$, $D_{\text{path}} = 0$, $\mathcal{R}_{\text{length}} = 0$ for every large λ in a first attempt. We therefore solve $\lambda = 0$ first, warm-start each subsequent λ from the previous solution, and verify explicitly that each reported point beats the trivial one. This is a general hazard for path-deviation objectives, not a quirk of this system.

	λ	F	D_{path}	$\mathcal{R}_{\text{length}}$
0 (fidelity only)	1.0000	0.459	56	
	0.2	0.9991	0.295	115
	0.5	0.9980	0.237	161
	1.0	0.9943	0.192	167
	2.0	0.9920	0.155	172
	5.0	0.878	0.075	94

Table 3: Reachability frontier for the pump family. Path-optimality is nearly free until $D_{\text{path}} \approx 0.155$ and expensive thereafter. A row at $\lambda = 10$ was removed: it is dominated by the trivial solution below, and is therefore not a Pareto point.

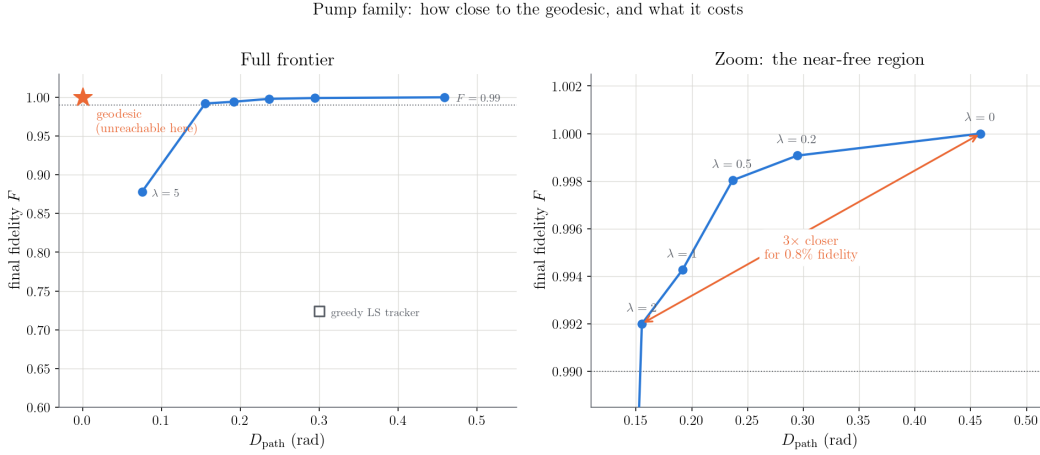


Figure 3: Fidelity against geodesic deviation for the pump family. The greedy least-squares tracker of Eq. (4) is dominated by the globally optimized frontier, confirming its single point was not the limit.

5.7 Gates approach the geodesic, but not as closely

Applying the same sweep to a four-layer displacement+SNAP sequence (Table 4) settles the comparison on equal footing: both families optimized against a geodesic-aware objective, compared at matched fidelity. Gates improve substantially on their fidelity-only value — 1.116 to 0.693 — but the curve flattens as fidelity begins to erode, approaching an asymptote near 0.7. Pulses reach 0.155 while still at $F = 0.992$.

At $F \approx 0.99$ the two families sit at $D_{\text{path}} \approx 0.16$ and $D_{\text{path}} \approx 0.76$ respectively: a factor of four to five. Notably this gap is *wider* than the fidelity-only comparison suggested (0.46 versus 1.03, a factor of 2.2). Testing both arms properly sharpened the distinction rather than dissolving it, and it is the velocity-deficit of Eq. (4) that supplies the mechanism.

5.8 A learned policy does not close the gap

For completeness we trained a PPO policy on a minimum-time formulation of the same problem (destination-only reward: per-step $-\Delta t$, terminal bonus, and potential-based shaping on the angle to the target). The environment was validated by replaying the optimized pulse, which reproduces $F = 0.9923$ exactly. From a random start the policy reaches $F = 0.593$ — the squeezing ceiling again — because the default exploration scale is roughly $20\times$ the perturbation that already destroys a good solution (1% amplitude noise costs 0.14 fidelity, 3% costs 0.55). Behaviour cloning from the optimized pulse reaches only 0.32 owing to covariate shift; noise-augmented cloning reaches 0.821, and PPO fine-tuning does not improve on it (0.818). Open-loop optimal control is already optimal on this deterministic problem, so a tie was the best available outcome and we did not achieve it.

	λ	F	D_{path}	$\mathcal{R}_{\text{length}}$
0 (fidelity only)	0.9964	1.116	20.1	
	0.10	0.9970	0.906	21.5
	0.20	0.9947	0.834	19.3
	0.35	0.9896	0.788	20.4
	0.50	0.9866	0.735	19.8
	0.80	0.9702	0.693	15.9

Table 4: Reachability frontier for a four-layer displacement+SNAP sequence. Every row beats the trivial solution of Section 5.6. Compare Table 3: at matched fidelity $F \approx 0.99$, pulses reach $D_{\text{path}} \approx 0.16$ where gates reach only ≈ 0.76 .

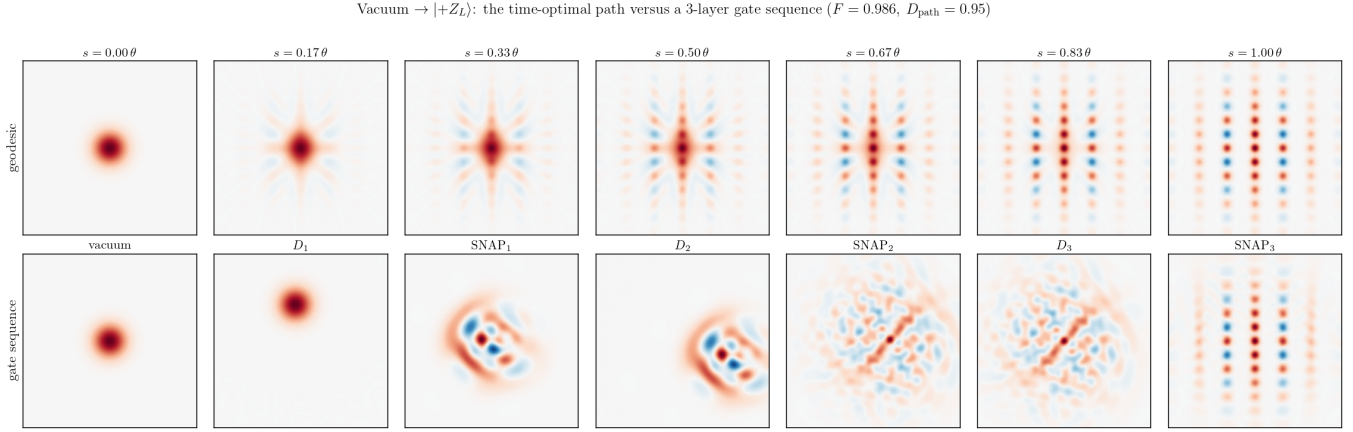


Figure 4: The time-optimal path (top) against a displacement+SNAP sequence (bottom) reaching comparable fidelity. This is the qualitative content of D_{path} .

6 Physics diagnostics

Numerical geometry should be checked against the physics it summarizes. Figure 4 places the Wigner function after each gate beside the geodesic sampled at equal arc length. Both rows begin at vacuum and end at the target, so everything in between is the protocol’s detour made visible: the geodesic interpolates cleanly and symmetrically between the two, while the gate sequence displaces the state far across phase space before reassembling the lattice.

7 Planned work

Geodesic waypoints as a fidelity objective.. The natural next experiment keeps everything in the language of fidelity. Rather than penalizing D_{path} , prescribe waypoints $s_1 < \dots < s_m$ along the geodesic and optimize

$$\mathcal{L} = - \sum_{j=1}^m w_j \left| \langle \gamma(s_j) | \psi(t_j) \rangle \right|^2, \quad (6)$$

so every term is an ordinary state-transfer fidelity. Two reasons to expect this to help: it decomposes a long-horizon credit assignment problem into short stages, much as the Δ -curriculum did; and it is physically transparent, since each target is a specific state whose Wigner function can be inspected. As with Section 5.5, this places the geodesic in the objective and therefore measures reachability.

ECD under the same test.. Section 5.7 settles the comparison for displacement+SNAP. Repeating it for ECD would establish whether the ≈ 0.7 asymptote is a property of gate-based control generally or of that one instruction set, and is the natural completion of Table 2.

Wigner diagnostics along every protocol.. Extending Figure 4 to ECD and to the waypoint-optimized sequences gives a qualitative check on whether “closer in D_{path} ” corresponds to “visibly cleaner and more symmetric” — the physical claim the metric is meant to formalize.

8 Limitations

Everything reported here concerns a single target, $|+Z_L\rangle$ at $\Delta = 0.3$ with $n_{\text{fock}} = 100$. Absolute times are nominal: the operators are subspace normalized, so only dimensionless ratios ($\mathcal{R}_{\text{length}}$, D_{path} , $\mathcal{R}_T$) are meaningful. $\mathcal{R}_T$ is additionally convention-sensitive, since the realized energy uncertainty is far below the family’s worst-case bound. All fidelities are multi-start local optima, not certified global ones; the reinforcement learning arm is a single seed at modest tuning effort. Two questions remain open for methodological rather than physical reasons: whether the continuous/discrete distinction survives a within-family test, and whether feedback helps under control noise, whose open-loop comparator overfit its training perturbations. Finally, the static baseline reported by the numerical layer was never reproduced exactly; the protocol details are not recorded in the shared repository.